Cryptography 101



presented by hugo, @meow on discord

DISCLAIMER:

IM NOT HERE TO TEACH YOU HOW TO MATH OR TO GOOGLE, IM HERE TO GET YOU SITUATED AND HOPEFULLY FIND SOME TOOLS THAT ARE USEFUL

the rest is up to you <3

Philosophy

- CryptoHack teaches better than me and that’s ok </3
- Why I write challenges?
  - writeups.

Im here to teach you Cryptography the CTF category

mostly little tricks i have picked up over the years

what is crypto?

- math



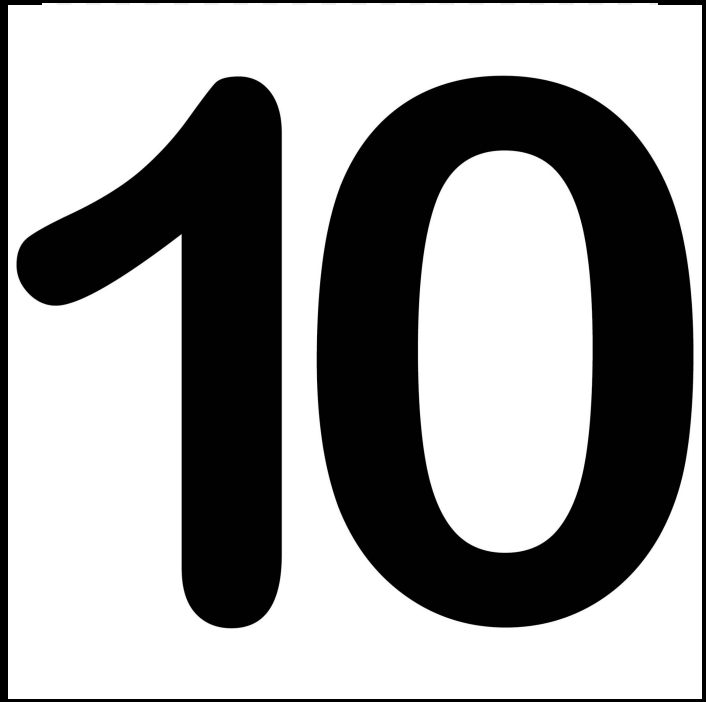
what is crypto?

- math
- gambling



what is crypto?

- math
- gambling
- big numbers



what is crypto?

- math
- gambling
- big numbers
- paying money to hetzner



what is crypto?

- math
- gambling
- big numbers
- paying money to hetzner
- ~~paying money to openai~~



General flow of a Crypto chall

Reverse style challenge

```
try:
    from secret import FLAG
except:
    FLAG = "friday{fake_flag}"
```

```
# Two primes are too boring to generate, if I take two consecutives, it just works
p = randprime(2**1023, 2**1024 - 1)
q = nextprime(p)

n = p * q
d = pow(e, -1, (p - 1) * (q - 1))
```

```
print("n = ", n)
print("CIPHER =", encrypt(flag, d, n))
```

General flow of a Crypto chall

Riddle me this style chall

```
print("1) explore:")
print("2) fight:")
m = input("> ").strip().lower()
```

```
print("Speak the magic word and save the world")
for _ in range(16):
    inp = input("Speak the magic word: ").strip()

    nonce, secret_phrase, magic_word = attempt(key, b"give me the flag", storm)

    if inp == magic_word:
        print("       [  Lain descends, the storm dissipating...  ]")
        with open("flag.txt", "r") as f:
            print(f.read())
        exit()
    else:
        print(f"The word was incorrect, maybe carried away by the harsh winds...
```

2 sorts of crypto!

Boomer

classical pre computer algorithms

^ in modern ctfs we call these riddles, because making it hard without being guessy is hard

vs lbh pna ernq guvf, lbher n areq

Clanker

computer assisted math

^ what we generally call crypto now, what we will spend most of our time on

theory!

prerequisites

it’s gonna be nice if you know at least a little bit of the following:

- probability
- modular arithmetic
- googling
- python

public key

also known as asymmetric crypto

publish a key, people send you messages with it

mostly gonna be RSA and ECDSA

aicc2 :3

private key

symmetric crypto

both people know a secret, use it to communicate

mostly fancy xor chains and various AES

what key?

hashes and pure maths

how do i solve this question without waiting for the heat death of the universe

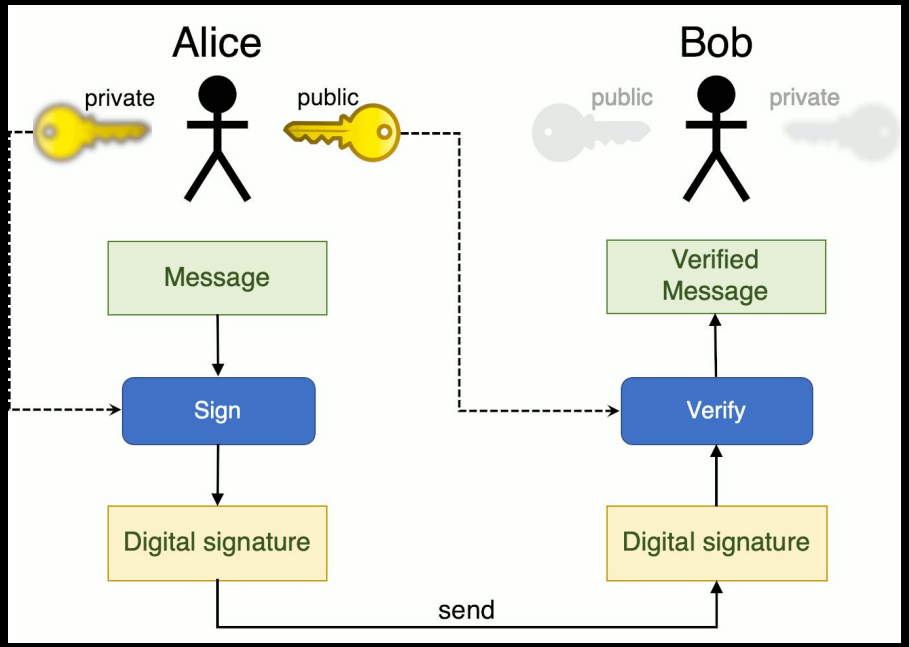
gambling!!!!!

public key/asymmetric key

Bob and Alice

- encryption & signature
- depends on which key you use
- used for communication mostly

generally wont see this out of crypto and maybe web challs



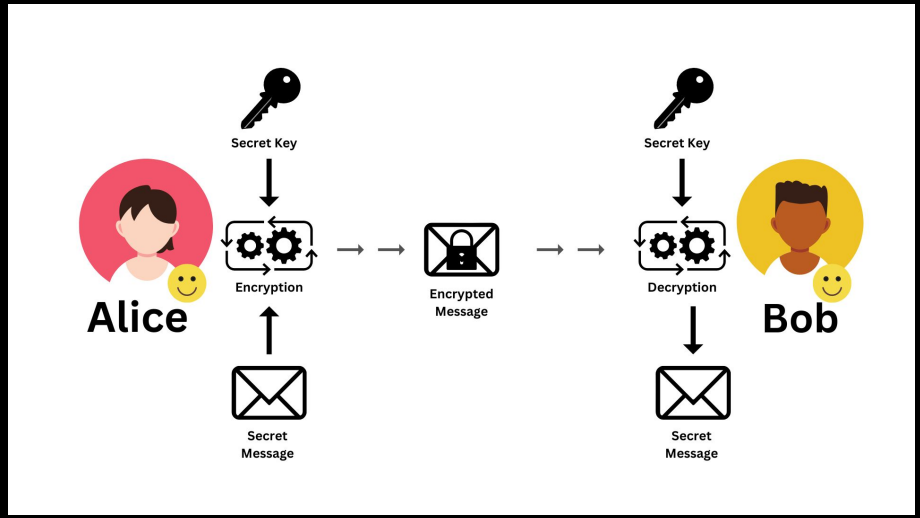
private key/symmetric key

both bob and alice start out knowing a secret!

most likely using an asymmetric key cipher to exchange keys

block ciphers, google it

tends to be relatively common in rev and pwn challs for some reason



the rest

- hashes
  - md5, sha
- RNG
  - pseudorandom numbers
- straight up maths lol
  - chinese teams love this

Hashes and RNG are useful everywhere, good to know how to mess with them



tools of the trade!

Google

- All crypto challs are either rev challs or osint challs
- any hard enough rev chall is an osint chall (stalk the author)
- any hard enough osint chall is a rev chall (fuck it im doing it myself)



cyberchef

- for boomer crypto
- good for non crypto players to play with things without having to know python
- also good for playing around with blind crypto, might allow you to gain some insight in how something works
- https://gchq.github.io/CyberChef/



python 3

- easy to write
- arbitrarily large integer handling by default
- pycryptodome, pwntools, numpy, sympy and other cool libraries!
- learn pycryptodome and never implement anything yourself



if youre on linux you might have to make a venv to handle all the packages youre gonna need for python, more info on the next slide

venv setup for dummies

1. python3 -m venv [name for your venv, usually .venv]
2. source [name]/bin/activate

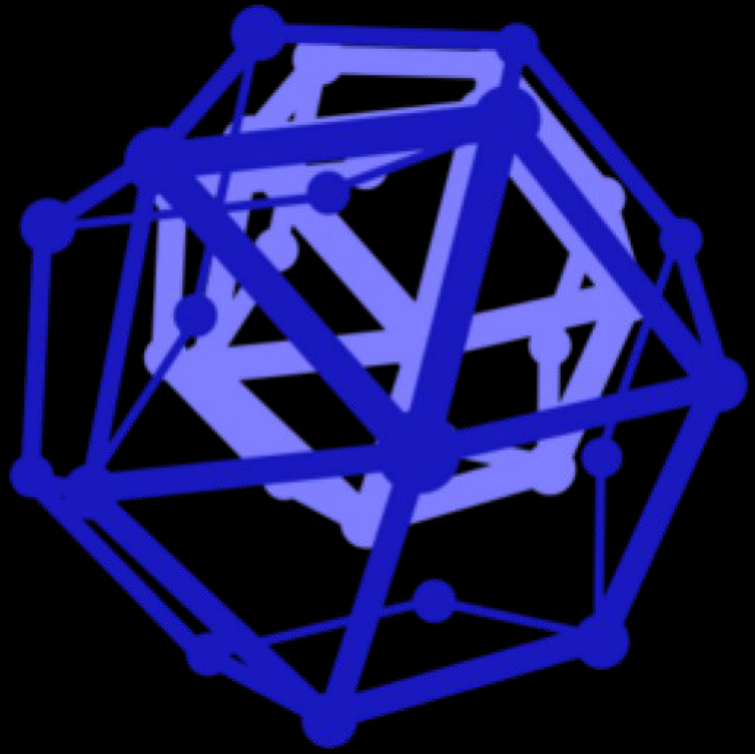
then you will be able to pip install whatever you want

to exit the venv, simply type:

source deactivate

sage

- python for nerds, a lot more handling for math related behaviour
- medium difficulty rev challenge to set up on most machines, sowwy (more info next slide)
- im kinda ass at it, but it seems most modern llms have been castrated in their ability to use sage lol
- beware of CVEs (xtea)



DO NOT TRY TO INSTALL SAGE YOURSELF, IT WILL ONLY CAUSE PAIN

Instead, install the CryptoHack docker, or make your own CTF Docker Setup.

- https://github.com/cryptohack/cryptohack-docker
- theres a talk by fl0w that gives a cool personal one he uses, look up how to get sage set up in there its also very good.
- https://discord.com/channels/681251412155564080/1158831244491108463/1357764498269929712

misc

- **rockyou.txt**: find this yourself, legality is questionable
- **johntheripper:** best password cracker
- **gmpy2**: python library, quite useful. use sage tho its better
- **dcode.fr**: for if you dont want to write boomer attacks
- **cryptohack docker**:
  https://github.com/cryptohack/cryptohack-docker
- **arXiv**: https://www.arxiv.org
- **iacr**: https://eprint.iacr.org
- **factordb**: https://www.factordb.com
- **magma computer algebra** ← only for giga nerds

Practical

- lets do some osint!
- feel free to try to stalk my github stuff, but try to osint the contents and not the chall itself, i know there are writeups already written for it that mention the paper.

[little example with Conversation, my chall from lake finals]

anyone who dms me (@meow) on discord with the paper before the end of the session can come claim a chocolate.

:3

questions?

PRNG lectures!

we’re gonna take a look at what authors like to do with some prngs. figure the maths out yourself on wikipedia i believe in u

you can go on to the challs if you’re comfortable reading yourself, we will be working on:

- lcgs
- lfsr
- mt
- learn xorshift yourself, you’ll need it for the challs

Important things

- beware of schizophrenia
- important tools
  - modular inverse, polynomials and resultants: learn this in class
  - lattice attacks: learn this with cryptohack
  - brute force: learn this with your credit card
- goal of breaking prngs:
  - predict the future
  - figure out the past

Linear Congruential Generators

simplest one

X = (a*X+c) % m, usually m is a power of 2

people love these because you get forced to learn maths

take a look at RBG+ and RBG++ for a more difficult chall based on this

ive been working on efficient resultant clearing for the juicer for MONTHS HELP MEEEEEEEE

Linear Feedback Shift Register

feed the outcoming bits back into it

weak to known plaintext attacks, we know whats in the register we can find the rest

MT

python uses this

magic number of 624

steal someone else’s work, or implement the matrix yourself im not your mom

friday challenges??? in this economy??

friday.polygl0ts.ch!!!!!